Chapter 2

Data, Its Types and Data Classification

Variable

A variable is a characteristic or value that can change from one person, object, or situation to another.

Example:
Suppose we have students:

Student	Age
Ali	20
Sara	21
Ahmed	19

Here, **Age** is a variable because its value changes from student to student.

Other examples of variables:

- Height
- Weight
- CGPA
- Marks
- Blood group
- Number of family members

Qualitative Variable

A **qualitative variable** describes a **quality or category**.
Its values are usually **words/categories**, not numerical measurements.

Examples:

- **Blood Group:** A, B, AB, O
- **Gender:** Male, Female
- **Eye Color:** Black, Brown, Blue
- **Department:** CS, IT, SE

Quantitative Variable

A **quantitative variable** represents something that can be **measured or counted using numbers**.

Examples:

- **Age:** 20, 21, 22
- **Height:** 170 cm, 175 cm
- **Weight:** 60 kg, 70 kg
- **Marks:** 80, 75, 90
- **Number of family members:** 4, 6, 8

Quantitative variables have two types

1. Discrete Variable
A variable that is **counted** and usually takes whole-number values.

Examples:

- **Number of students:** 50
- **Number of family members:** 6
- **Number of cars:** 3

You normally cannot have 3.5 family members.

2. Continuous Variable

A variable that is **measured** and can take decimal values.

Examples:

- **Height:** 175.5 cm
- **Weight:** 65.7 kg
- **Temperature:** 36.5°C

EXAMPLE 2.2 Variables and Data

Human Blood Types Human beings have one of four blood types: A, B, AB, or O. What kind of data do you receive when you are told your blood type?

Solution Blood type is a qualitative variable because its possible values are non-numerical. Therefore your blood type is qualitative data.

EXAMPLE 2.3 Variables and Data

Household Size The U.S. Census Bureau collects data on household size and publishes the information in *Current Population Reports*. What kind of data is the number of people in your household?

Solution Household size is a quantitative variable, which is also a discrete variable because its possible values are 1, 2, ... Therefore the number of people in your household is discrete, quantitative data.

EXAMPLE 2.4 Variables and Data

The World's Highest Waterfalls The *Information Please Almanac* lists the world's highest waterfalls. The list shows that Angel Falls in Venezuela is 3281 feet high, or more than twice as high as Ribbon Falls in Yosemite, California, which is 1612 feet high. What kind of data are these heights?

Solution Height is a quantitative variable, which is also a continuous variable because height can conceptually be any positive number. Therefore the waterfall heights are continuous, quantitative data.

Study vs Auxiliary Variable

1. Study Variable

The study variable is the main variable that you are interested in studying or analyzing.

Example:
Suppose you want to study the average weight of students.

- Weight: **Study variable**

2. Auxiliary Variable

An **auxiliary variable** is an additional variable that helps you understand, estimate, or analyze the study variable more effectively.

Example:

Suppose you are studying students' weight:

- Weight: **Study variable**
- Age, height, gender: **Auxiliary variables**

Data

Data is a collection of facts, information, or observations that are collected for a specific purpose.

Examples:

- **Students' marks:** 70, 85, 60, 90
- **People's ages:** 20, 25, 30
- Names, genders, heights, temperatures, sales, etc.

Types of Data

1. Primary Data

Primary data is the data that is collected directly by the researcher for a specific purpose.

Examples:

- Conducting a survey
- Interviewing people
- Taking measurements

- Doing an experiment

2. Secondary Data

Secondary data is data that has **already been collected by someone else** and is used by a researcher.

Examples:

- Government reports
- Books and research papers
- Websites
- Existing company records

3. Quantitative Data
Data that is represented by numbers and can be counted or measured.

Examples:

- **Age:** 20 years
- **Height:** 175 cm
- **Marks:** 85
- **Weight:** 70 kg
- **Number of family members:** 5

2. Qualitative Data
Data that represents **categories or qualities** and is usually expressed in words.

Examples:

- **Gender:** Male/Female
- **Blood Group:** A/B/AB/O
- **Eye Color:** Brown/Black
- **Department:** CS/IT/SE

Classifications of data

Data representation means showing or organizing data in a clear and understandable form so that it can be easily analyzed and interpreted.

1. Time Series Data

Time series data is data collected or recorded **at different points in time**, usually in chronological order.

Here, the same unit is observed repeatedly, while the time changes.

Example: **Temperature of Lahore**

Date	Temperature
Monday	30°C
Tuesday	32°C
Wednesday	31°C
Thursday	33°C

2. Cross-Sectional Data

Cross-sectional data is data collected from **different people, places, or groups at the same point in time**.

Here, the **units change**, while the time remains the same.

Example: **Temperature of different cities on Monday**

City	Temperature
Lahore	30°C
Karachi	32°C
Islamabad	25°C
Peshawar	28°C

3. Panel Data

Panel data combines the features of **time series data and cross-sectional data**.

It collects data from multiple units over multiple points in time.

Example: Temperature of different cities over several days

Date	Lahore	Karachi	Islamabad
Monday	30°C	32°C	25°C
Tuesday	32°C	33°C	26°C
Wednesday	31°C	31°C	24°C